KEY TOPIC 3 • PUPIL WORKBOOK

Medicine in 18th & 19th Century Britain

c1700–c1900: Ideas on Causes, Prevention, Hospital Care, Surgery & Public Health



Primary Visual Evidence: James Gillray, *The Cow-Pock—or—the Wonderful Effects of the New Inoculation!* (1802 satire of Edward Jenner)

HISTORICAL SATIRE ARCHIVE

PUPIL NAME: _____ TARGET: _____ CLASS: _____

KEY TOPIC 3 SPECIFICATION FRAMEWORK (1HI0/11)	LESSON DONE	QUIZ [10]
Key Topic 3.1: Ideas on Causes: Pasteur's Germ Theory & Koch's Bacteriology *How did swan-neck flasks (1861) and agar cultures (1876–83) demolish spontaneous generation?*	☐	☐
Key Topic 3.2: Approaches to Prevention: Edward Jenner & The Smallpox Vaccine *How did the 1796 cowpox experiment defeat variolation, and why did opposition persist until 1852?*	☐	☐
Key Topic 3.3: Improvements in Hospital Care: Florence Nightingale & Professional Nursing *How did Scutari (1854–56), Notes on Nursing (1859), and the pavilion plan transform Victorian mortality?*	☐	☐
Key Topic 3.4: The Surgical Revolution: Simpson's Chloroform & Lister's Antiseptics *How did Simpson conquer pain, Lister conquer infection, and the 1890s introduce aseptic surgery?*	☐	☐
Key Topic 3.5: Public Health & Cholera: John Snow & The 1875 Public Health Act *How did John Snow's Broad Street pump map (1854) and the 1875 Act permanently dismantle government laissez-faire?*	☐	☐

Living Timeline • Part 1: Vaccines, Anaesthetics & Early Sanitation (1700–1854)

Living Timeline Dual-Coding Protocol: Complete each milestone sketchpad using dual-coding (combining visual storytelling symbols, causal arrows, and dates) to master the chronological spine across the centre-fold.

1796 • Edward Jenner: Smallpox Vaccination & Immunology

Key Topic 3.2

Jenner inoculates James Phipps with cowpox pus from Sarah Nelmes, proving cowpox confers immunity against smallpox. Publishes findings in 1798. Despite initial Royal Society scepticism, vaccination eradicates inoculator infection risks and founds immunology.

1847 • James Young Simpson: Chloroform & Anaesthesia

Key Topic 3.4

Simpson discovers the potent anaesthetic properties of chloroform in Edinburgh, ending agony during surgery. Queen Victoria's use of chloroform during childbirth (1853) popularises it, enabling complex operations despite the initial 'Black Period' of surgery.

1848 • First Public Health Act & The General Board of Health

Key Topic 3.5

Spurred by Chadwick's 1842 report and epidemic cholera, Parliament passes the permissive 1848 Act. Towns can create local boards of health to supply clean water and drainage, though lack of compulsion leaves most cities vulnerable to filth and contagion.

1854 • John Snow: Broad Street Pump & Waterborne Cholera

Key Topic 3.5

During the Soho epidemic, Snow compiles his famous spot map linking 500+ fatal cholera cases to the Broad Street pump. Removing the handle immediately halts the outbreak, mathematically disproving the miasma theory seven years before Pasteur's Germ Theory.

The 18th century relied on empirical observation; the 19th century unlocked the laboratory.

Living Timeline • Part 2: Nursing, Microbes & Antiseptics (1854–1875)

Living Timeline Dual-Coding Protocol: Draw visual symbols for Crimean hospital sanitation, swan-neck flasks, Lister carbolic sprays, and municipal sewers across the timeline below.

1854–56 • Florence Nightingale: Hospital Sanitation at Scutari

Key Topic 3.3

During the Crimean War, Nightingale enforces rigorous cleanliness, ventilation, and fresh nutrition at Scutari hospital, slashing soldier death rates from 40% to 2%. Returns to Britain to establish the Nightingale Training School and author Notes on Nursing (1859).

1861 • Louis Pasteur: Germ Theory & Swan-Neck Flask Experiments

Key Topic 3.1

Commissioned by Lille brewers, Pasteur disproves spontaneous generation using swan-neck flasks, demonstrating that airborne microbes cause fermentation and decay. In 1861, he publishes Germ Theory, proving microscopic germs cause specific diseases in living organisms.

1865–67 • Joseph Lister: Carbolic Acid Spray & Antiseptic Surgery

Key Topic 3.4

Applying Pasteur's Germ Theory to wound sepsis, Lister uses carbolic acid sprays, dressings, and hand-washing during surgery at Glasgow Infirmary. His antiseptic techniques reduce amputee mortality from 46% to 15%, paving the way for modern aseptic operating theatres.

1875 • The Second Public Health Act: End of Laissez-Faire

Key Topic 3.5

Disraeli's government ends laissez-faire by making public health compulsory. Municipal councils are legally required to provide clean piped water, subterranean sewer drainage, street lighting, and medical officers of health, transforming urban life expectancy.

Why did it take 30 years for Germ Theory to transform medical understanding in Britain?

Core Specification Focus: Germ Theory (1861); Louis Pasteur; Robert Koch; bacteriology; agar cultures, chemical dyes, and microscopes; impact on Britain.

1 c1700–1850s • Why did doctors believe disease created germs, rather than vice versa? [1.1]

• Smelling rotting meat and open cesspools	
• Foul airborne damp blamed for illness	
• Microscopes seeing tiny swimming creatures	

2 1861 • How did the swan-neck flask disprove spontaneous generation? [1.2]

• Heating sour beer inside glass flasks	
• Bent swan necks trapping airborne dust	
• Boiled broth staying clear and fresh	

3 1867–1876 • Why did British doctors refuse to believe germs caused human illness? [1.3]

• London doctors defending bad air theory	
• Shining light beams through dusty boxes	
• Spraying smelly carbolic acid on cut flesh	

4 1876–1882 • How did Koch's staining and culture methods prove specificity? [1.4]

• Solid seaweed agar jelly in shallow dishes	
• Staining microbes with bright purple dyes	
• Photographing deadly tuberculosis under glass lenses	

5 1883–1890 • Why was photomicrography crucial in winning British doctors over? [1.5]

• Tracking curved cholera germs in river water	
• Taking crisp black-and-white photos of germs	
• Injecting farm animals to test poisons	

Pasteur proved that rotting soup doesn't spontaneously create germs; Koch proved which specific germs caused tuberculosis.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why there was rapid progress in understanding the causes of disease in the period c1860–c1890. [12 marks]

Stimulus: **Louis Pasteur's Germ Theory (1861)** **Robert Koch's work on bacteria** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

swan-neck flask • Germ Theory (1861) • spontaneous generation • Robert Koch • agar jelly • methyl violet dye • tuberculosis (1882) • microphotography

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 3.1). In the drawing box, sketch Pasteur's swan-neck flask experiment and Koch's agar plate. Annotate how identifying specific bacteria ended miasma theory forever! **[ ] Done**

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** SCORE: [/ 4] • 5 MINS

Explain one way in which ideas about the cause of disease in the late nineteenth century (c1860–c1890) were different from ideas in the Renaissance (c1500–c1700). [4 marks]

Contrast laboratory discoveries of specific microbes with earlier beliefs in miasma and spontaneous generation.

Structure your causal argument: demonstrate how Koch built directly upon Pasteur's Germ Theory to revolutionize diagnosis.

Why did Edward Jenner face fierce resistance despite discovering the smallpox vaccine?

Core Specification Focus: Edward Jenner; development of smallpox vaccination (1796); opposition from doctors, church, and variolators; government funding and compulsory vaccination (1852).

1 1721–1795 • Why was inoculation dangerous for poor urban communities? [2.1]

- **Scabbing sores blinding young children**
- **Scratching live pus into healthy arms**
- **Paying rich doctors high guinea fees**

2 1796–1798 • How did Jenner use scientific experimentation to test folklore? [2.2]

- **Milking cows with clear smooth skin**
- **Scraping blister liquid into boy James Phipps**
- **Dosing boy with deadly pox without illness**

3 1798–1810s • Why did London doctors and church leaders resist vaccination? [2.3]

- **Printing scary cartoons with sprouting cow horns**
- **Church pulpits warning against beast blood**
- **Dirty hospital needles passing deadly infections**

4 1802–1840 • Why did the British government abandon laissez-faire for vaccination? [2.4]

- **Voting thirty thousand pounds in gold coins**
- **Giving free scratches to village babies**
- **Banning old dangerous pus cuts by law**

5 1852–1871 • Why did making vaccination compulsory spark fierce public protests? [2.5]

- **Fining poor parents for refusing jabs**
- **Angry street protests marching with banners**
- **Plunging death rates as smallpox disappears**

Cartoons showed vaccinated people growing cow heads and horns: proof that anti-vax memes existed long before social media.

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

EXTENDED WRITING ASSESSMENT

'Edward Jenner's development of the smallpox vaccine was the most significant breakthrough in the prevention of disease in the period c1700–c1900.' How far do you agree? [16+4 marks]

Stimulus: **Inoculation** **The Public Health Act (1875)** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

On the one hand, a major breakthrough was... • For example, ... • This was significant because... • On the other hand, an alternative factor was... • In particular, ... • Overall, the most decisive factor was... because...

KEY VOCABULARY BANK:

Edward Jenner (1796) • cowpox • James Phipps • smallpox • variolation • inoculation • compulsory vaccination (1852) • Anti-Vaccination League

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 3.2). In the drawing box, sketch Jenner inoculating James Phipps with cowpox. Annotate how this empirical experiment laid the foundation for modern immunology! **[ ] Done**

Jenner saved millions with cowpox, yet died before anyone could explain what a virus actually was.

Continue your response to Question 5/6 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which methods of preventing smallpox in the late eighteenth century were different from methods used in the seventeenth century. [4 marks]

Contrast safe cowpox vaccination with earlier reliance on prayer, quarantine, and risky variolation.

How did Florence Nightingale transform hospitals from death traps into places of cure?

Core Specification Focus: Florence Nightingale; hospital conditions at Scutari (1854–56); Notes on Hospitals and Notes on Nursing (1859); pavilion hospital plan; professional nursing training.

1 c1700–1850 • Why were early Victorian hospital wards notorious death traps? [3.1]

- **Crowded dark rooms smelling of festering wounds**
- **Drunk ward women sleeping on dirty rags**
- **Sharing blood-soaked blankets between beds**

2 1854–1856 • How did Nightingale's sanitary reforms drastically cut the death rate? [3.2]

- **Sewers overflowing beneath wooden ward floors**
- **Scrubbing dirty walls with boiling water and soap**
- **Feeding wounded soldiers hot fresh stew**

3 1859 • Why did Nightingale's belief in miasma still produce effective hygiene? [3.3]

- **Opening high windows for fresh river breezes**
- **Drawing colourful polar pie charts for MPs**
- **Demanding clean white cotton bandages**

4 1860s • How did the pavilion plan prevent cross-infection between wards? [3.4]

- **Building long narrow brick ward pavilions**
- **Washing smooth glazed walls with water**
- **Cross-ventilation blowing stale room air away**

5 1860–1890s • How did nurse training permanently raise the standard of hospital care? [3.5]

- **Uniformed women learning medical rules at St Thomas'**
- **Stern matrons inspecting spotless bedside tables**
- **Fusing clean hands with microbe science**

Nightingale believed miasma caused disease, yet her obsession with fresh air, scrubbing, and clean drains saved thousands of lives.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why the standard of hospital care and nursing improved so significantly in the second half of the nineteenth century. [12 marks]

Stimulus: **Florence Nightingale's work in the Crimea (1854–56)** | **The pavilion hospital plan** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

Florence Nightingale • Scutari (1854) • Notes on Nursing (1859) • pavilion plan • cross-ventilation • St Thomas' Hospital • trained nurses • sanitation

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 3.3). In the drawing box, sketch the Pavilion Hospital plan with its separate, ventilated pavilions. Annotate how cross-ventilation prevented hospital gangrene! **[ ] Done**

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which hospital care in the late nineteenth century was different from hospital care in the Medieval period. [4 marks]

Contrast modern sanitary pavilion wards and trained medical care with medieval monastic care.

Why did the conquest of pain in surgery temporarily increase the patient death rate?

Core Specification Focus: Surgical breakthroughs: pain (anaesthetics: nitrous oxide, ether, James Simpson and chloroform); infection (Joseph Lister and carbolic acid; asepsis); bleeding.

1 c1700–1846 • Why was surgical speed essential before the discovery of anaesthetics? [4.1]

- **Holding down screaming patients with leather straps**
- **Sawing off broken legs in thirty seconds**
- **Breathing sweet laughing gas at fairgrounds**

2 1847–1853 • Why did the death of Hannah Greener cause panic among surgeons? [4.2]

- **Sniffing heavy liquid vapour under dinner table**
- **Young Hannah Greener dying from cold overdose**
- **Queen Victoria giving birth with zero pain**

3 1846–1865 • Why did conquering surgical pain lead to higher patient death rates? [4.3]

- **Wearing stiff dried blood coats as badges**
- **Cutting deeper into belly flesh for longer**
- **Black rotting gangrene killing feverish patients**

4 1865–1870s • How did Lister apply Germ Theory to solve hospital gangrene? [4.4]

- **Treating green sewage water with smelly chemicals**
- **Wrapping boy Jamie Greenlees' fractured leg**
- **Pumping fine acid mist over open surgical wounds**

5 1880s–1890s • What is the crucial difference between antiseptic and aseptic surgery? [4.5]

- **Boiling metal scalpels in pressurized steam kettles**
- **Pulling on thin sterile rubber gloves**
- **Tying white cotton masks over mouths and noses**

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why surgery became significantly safer in the period c1840–c1900. [12 marks]

Stimulus: **James Simpson's discovery of chloroform (1847)** | **Joseph Lister's use of carbolic acid (1865)** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

James Simpson • chloroform (1847) • Black Period • Joseph Lister • carbolic acid spray (1865) • antiseptics • steam sterilisation • aseptic surgery

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 3.4). In the drawing box, sketch Lister's carbolic donkey engine spraying mist over an open wound. Annotate the difference between antiseptic and aseptic surgery! **[ ] Done**

Surgeons used to take pride in blood-caked frock coats; Lister forced them to wash their hands and operate in carbolic acid mist.

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which surgical operations in the early nineteenth century (before 1846) were similar to surgery in the Medieval period. [4 marks]

Focus on raw surgical speed and physical restraint before effective chemical anaesthetics were discovered.

Why did it take 20 years for John Snow's cholera discovery to change government policy?

Core Specification Focus: Public health: cholera epidemics; John Snow and Broad Street pump (1854); public health acts (1848 permissive vs 1875 compulsory); Bazalgette and London sewers.

1 1831–1848 • Why did towns refuse to set up health boards under the 1848 Act? [5.1]

- **Violent watery sickness turning human skin blue**
- **Piles of human excrement rotting near houses**
- **Parliament passing optional town health boards**

2 1854 • How did Snow use epidemiological mapping to prove water transmission? [5.2]

- **Plotting black death bars on street map**
- **Brewery men drinking ale staying completely healthy**
- **Unscrewing heavy iron pump handle with wrench**

3 1854–1858 • Why did the medical establishment refuse to accept Snow's water theory? [5.3]

- **Cracked brick cesspool leaking into drinking well**
- **Medical journals laughing at dirty water theory**
- **Doctors burning pitch barrels in foggy streets**

4 1858–1865 • Why did fear of miasma paradoxically save thousands from cholera? [5.4]

- **Boiling brown river mud halting Parliament debates**
- **Soaking courtroom window curtains in lime juice**
- **Digging thousand miles of underground brick tunnels**

5 1875 • Why was the 1875 Act the definitive death blow to laissez-faire? [5.5]

- **Working factory men winning parliamentary votes**
- **Forcing town councils to pipe clean fresh water**
- **Hiring official medical inspectors for bad food**

Snow removed the Broad Street pump handle in 1854, proving cholera was water-borne seven years before Germ Theory.

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

'The work of John Snow was the main reason for improvements in public health in the nineteenth century.' How far do you agree? [16+4 marks]

Stimulus: Edwin Chadwick's 1842 Report | The Public Health Act (1875) *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

On the one hand, a major breakthrough was... • For example, ... • This was significant because... • On the other hand, an alternative factor was... • In particular, ... • Overall, the most decisive factor was... because...

KEY VOCABULARY BANK:

John Snow (1854) • Broad Street pump • waterborne cholera • Edwin Chadwick (1842) • Great Stink (1858) • Joseph Bazalgette • Public Health Act (1875) • laissez-faire

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 3.5). In the drawing box, sketch John Snow's Broad Street pump map and Bazalgette's sewer network. Annotate how cholera forced Parliament to abandon laissez-faire! [ ] Done

Continue your response to Question 5/6 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which government action on public health in the late nineteenth century was different from government action in the Renaissance. [4 marks]

Contrast compulsory public health legislation with earlier temporary emergency measures.

KEY TOPIC 3 NOTES & EXAMINATION STRATEGY

Notes: Record key causal links, essay plans, and teacher feedback for this unit.

PUPIL NAME: _____ TARGET: _____ CURRENT GRADE: _____

ENQUIRY	ASSESSMENT COMPONENT FOCUS	PAGE	SCORE	TEACHER COMMENT & NEXT STEPS
KT3.1	**Q4:** Explain Why Rapid Progress in Causes c1860–c1890	p. 5–6	/ 12	
	Q3: Explain One Difference (Causes: c1700 vs c1900)	p. 6	/ 4	
KT3.2	**Q5/6:** Evaluative Essay on Jenner Smallpox Vaccine c1700–c1900	p. 8–9	/ 20	
	Q3: Explain One Difference (Smallpox Prevention Methods)	p. 9	/ 4	
KT3.3	**Q4:** Explain Why Hospital Care & Nursing Improved c1850–c1900	p. 11–12	/ 12	
	Q3: Explain One Difference (Hospital Care: Medieval vs c1900)	p. 12	/ 4	
KT3.4	**Q4:** Explain Why Surgery Became Safer c1840–c1900	p. 14–15	/ 12	
	Q3: Explain One Similarity in Surgery (Medieval vs Pre-1846)	p. 15	/ 4	
KT3.5	**Q5/6:** Evaluative Essay on John Snow & 1875 Public Health Act	p. 17–18	/ 20	
	Q3: Explain One Difference in Public Health Action (1848 vs 1875)	p. 18	/ 4	
TOTAL	**Key Topic 3 Cumulative Examination Assessment Portfolio (10 Tasks)**	p. 4–18	/ 88	**Overall Grade:** _____ Teacher Target: _____

• DIGITAL LEARNING & RETRIEVAL HUB • INTERACTIVE LESSONS & QUIZZES

Interactive Textbook • 10 Questions Per Enquiry

KT3.1	KT3.2	KT3.3	KT3.4	KT3.5
Pasteur & Koch	Edward Jenner	F. Nightingale	Simpson & Lister	Snow & 1875 Act
LESSON HUB	LESSON HUB	LESSON HUB	LESSON HUB	LESSON HUB
PRACTICE QUIZ	PRACTICE QUIZ	PRACTICE QUIZ	PRACTICE QUIZ	PRACTICE QUIZ
Score: [/ 10]	Score: [/ 10]	Score: [/ 10]	Score: [/ 10]	Score: [/ 10]

Sanitation, antiseptics, and vaccinations conquered disease; retrieval practice will conquer your GCSE exam.